Low-Cost Soil Moisture Sensing for Smallholder Irrigation

Prepared by the Field Systems Group of the Department of Agricultural Engineering.

Smallholder farmers often irrigate on a fixed calendar because commercial soil moisture sensors cost more than a season's income from a small plot. We tested a coated circuit-board probe that costs under five dollars to build. Over one growing season it stayed within 2.5 percent of a reference neutron probe.

Background

Cheaper sensors exist, but their readings drift as the probe corrodes in wet soil. Drift forces farmers to recalibrate often, and few of them have the tools to do so.

Methods

We built 40 capacitive probes and compared them with a reference probe over one season.

1. Probes were buried at 15 cm and 30 cm depth in four plots.
2. Readings were logged every hour by a battery-powered board.
3. Each probe was recalibrated against soil cores every four weeks.

Results

The coated probes stayed within 2.5 percent of the reference throughout the season.

Table 1. Error against the reference probe

Sensor	RMSE (%)	Cost (USD)
Coated board probe	2.1	4.50
Bare board probe	4.8	3.20
Commercial probe	1.6	62.00

Conclusions

A coated probe costing under five dollars is accurate enough to schedule irrigation for smallholders. Field trials with 200 farms will start in March 2027.

Key findings

- Coating cut drift from 3.1 to 0.4 percent per month.
- The probes survived a full season without failures.
- Cost per sensor was 7 percent of the commercial unit.

Next steps

- Fit solar-powered radio links.
- Test in sandy and clay soils.
- Publish the design files openly.

Poster details

Poster ID	P-217
Session	Wednesday 14:00 to 15:30
Board	31
Contact	agritech-posters@eastbrook.example

References

[1] Field Systems Group (2024). Annual report on sensor drift. Eastbrook University Press.

[2] Irrigation Standards Board (2021). Guidelines for soil moisture measurement, second edition.

[3] Agri-Tech Research Day (2025). Proceedings of the poster session on low-cost instrumentation.

Acknowledgements

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